

LOYIHA BO'YICHA TO'LIQ TAQDIMOT

United Lane System

Logistika uchun yagona operatsion platforma: Fuel Service, Safety, Statistics, Driver va Admin bitta ish tizimida.

5

rolga asoslangan ish zonalari

12+

asosiy API yo'nalishlari

10

taqdimot sahifalari

- Frontend: React + Vite, moslashuvchan workspaces, xaritalar va jadvallar.
- Backend: FastAPI + SQLAlchemy + JWT, PostgreSQL/SQLite, Motive refresh.
- Integratsiyalar: Motive, TomTom, AI/OpenRouter, Telegram bot, Railway, Netlify.



Mahsulot xaritasi

Har bir user o'z roliga mos workspacega kiradi, backend esa API ruxsatlarini tekshiradi.

01 Admin

- Users, roles, bans
- Password resets va activity overview
- Oxirgi admin himoyasi

02 Fuel Service

- Loads, routing, tracking
- Fuel approvals va reconciliation
- Profitability, Full Road, Chat

03 Statistics

- Truck analytics
- Motive snapshots va archive trends
- MPG, mileage, faults, fuel

04 Safety

- Fleet safety, notes, documents
- Investigations va shift briefs
- AI chat, service/emergency tools

05 Driver

- Mobile-first kabinet
- Motive truck bilan bog'langan
- Fuel, route, service, chat

SHARED SYSTEM

Yagona kirish

- Auth pageda department tanlanadi
- Office accounts Admin orqali yaratiladi
- Driver truck match orqali ro'yxatdan o'tadi

Yagona backend

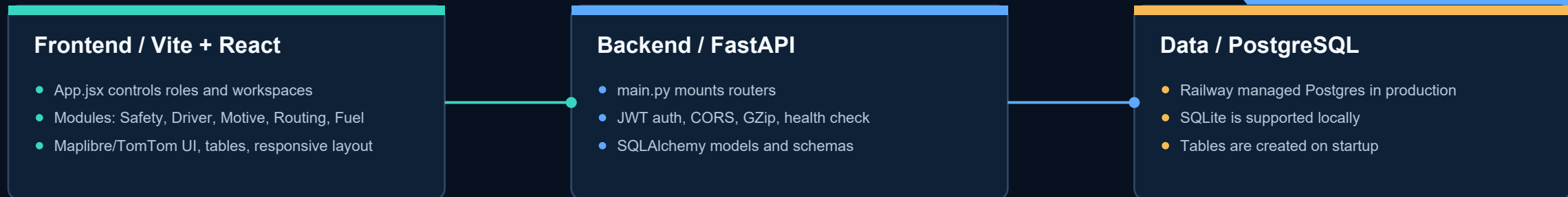
- Barcha routes /api ostida
- JWT access token
- department checks va ban checks

Yagona data layer

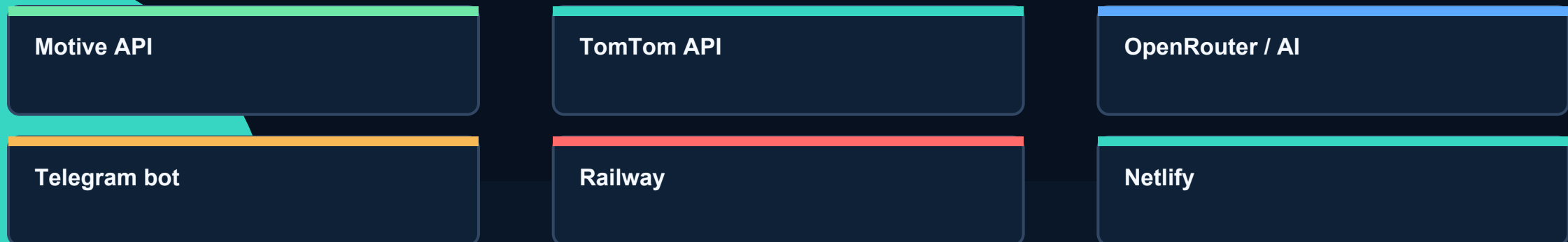
- Motive snapshots, loads, route history
- fuel authorizations, safety docs
- team chat va activity events

Tizim arxitekturası

Backend va frontend alohida bo'lgan monorepo: local development, Railway API, Netlify UI va PostgreSQL.



INTEGRATIONS AND RUNTIME



Fuel Service: asosiy operatsion markaz

Dispatch va fuel coordinators uchun workspace: loads, maps, approvals va lane margin analysis.

01 Dashboard

- Load counts, active loads, low fuel, review items, miles left

02 Loads

- Dispatch table: driver, truck, status, MPG, fuel %, pickup, stops, delivery, pay inputs

03 Tracking

- Motive fleet data: status, driver, location, movement, stale data, low fuel, faults

04 Routing

- Route A -> B, live truck fuel/location, branded fuel stops

05 Approvals

- gallons, amount, price cap, expiration, sent/used/violated, reconciliation

06 Team + Settings

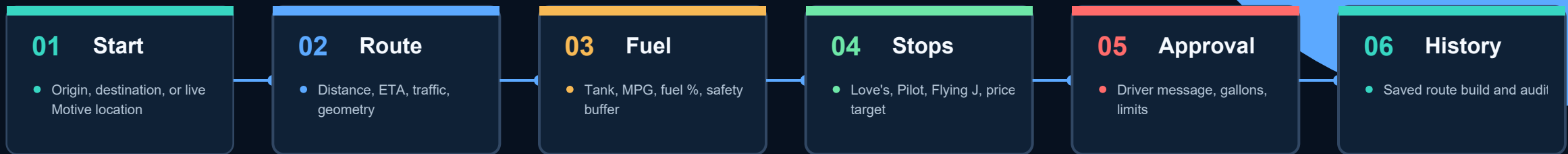
- Team Chat, workspace theme, font/background controls

FUEL WORKSPACE TABS

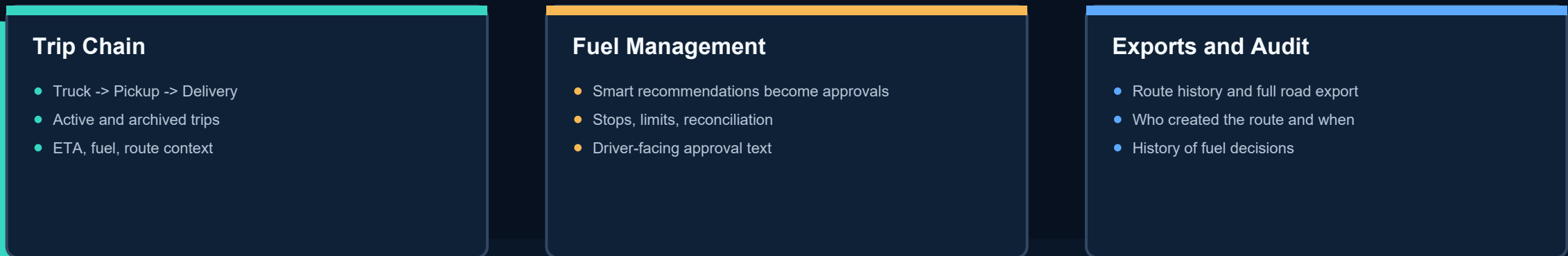
[Dashboard](#)[Tracking](#)[Statistics](#)[Profitability](#)[Full Road](#)[Routing](#)[Route History](#)[Approvals](#)[Loads](#)[Team Chat](#)[Settings](#)

Route operations flow sifatida quriladi

Tizim truck, origin, destination, fuel status, station data va approval workflowni bog'laydi.



FULL ROAD WORKSPACE



Statistics: sana, telemetriya va trendlar

Analytics Motive snapshots, saqlangan history va data date aniq ko'rsatishga tayanadi.

Fleet

Total / located / online vehicles

Status

Moving, stopped, stale states

Truck

MPG, fuel %, odometer, mileage

Risk

Faults, incidents, driver context

Trends

Daily stats and archive growth

Analytics View

- Fleet dashboard with total, located, moving, stopped, online, stale, active drivers
- Truck detail: location, telemetry, driver, fuel, speed, odometer
- Filters by moving, stopped, stale, low fuel, faults, truck/driver/location

Data Freshness

- Dates and timestamps stay near data so users know freshness
- Motive background refresh lowers wait time
- Exports and snapshots help after-shift review

Safety: nazorat, hujjatlar va yordam

Safety workspace compliance, fleet safety, service/emergency tools va AI supportni birlashtiradi.

01 Safety Notes

- Private notes for safety users
- Saved on backend

02 Documents

- Upload, category, summary
- Issues, actions, excerpt, AI analysis

03 Investigations

- Facts, evidence, questions
- Owner, due date, severity, outcome

04 Shift Briefs

- Handoff notes
- Checklist, action items, shift status

05 Service Tools

- Service centers and emergency options
- Driver/Safety support map

06 Safety AI Chat

- Operational safety questions
- Image attachments where supported

EXPORTS / HANDOFF / SERVICE

Investigations export, shift brief export, risky people export, Motive-based safety fleet snapshot, and emergency/service tools for Safety and Driver.

Odamlar va kundalik ish

Project access control, driver mobile flow va umumiy communicationni yopadi.

Driver Workspace

- Mobile-first interface
- Truck number + Motive vehicle match
- Route, fuel, service, emergency support
- No extra Admin/Fuel access

Admin Panel

- Create/edit Admin, Fuel, Statistics, Safety, Driver accounts
- Ban/unban, reset passwords, protected deletes
- Overview: users, routes, fuel auth, team messages, safety docs

Team Chat

- Shared rooms for departments
- Reply, edit own messages, delete own messages
- Safety/Admin moderation and search

AUTH PAGE AND UX

Appearance Controls

- Auth page has font/background settings

Loading UX

- Centered larger loaders show when data is loading

Backend qoidalar va tarixni saqlaydi

Muhim cheklovlar backendda: auth, roles, bans, persistence, refresh va integrations.

Security Model

- JWT authentication for protected routes
- Password hashes, not plain passwords
- Department checks and banned-user blocks

Data Models

- Users, activity events, loads, planner items
- Motive snapshots, daily stats, api connections
- Routes, full road trips, fuel authorizations, safety docs, chat

Runtime Config

- DATABASE_URL, SECRET_KEY, CORS_ORIGINS
- MOTIVE/TOMTOM/API/AI env variables
- Snapshot TTL, background refresh, gzip, workers

DEPLOY PATH

1. Railway DB

- Managed PostgreSQL
- DATABASE_URL reference

2. Railway API

- backend/ root
- FastAPI + worker/bot

3. Netlify UI

- frontend build
- VITE_API_URL=/api

4. Health

- /api/health
- docs at /docs

Project ichida nimalar bor

United Lane System to'liq private operations platform sifatida ishlashga tayyor.

Operational Value

- Less switching between sheets, maps, Motive, and chats
- Fuel decisions connect to route, limits, and reconciliation
- Safety and Driver get tools without extra permissions

Technical Base

- React/Vite frontend, FastAPI backend, SQLAlchemy persistence
- PostgreSQL production path, SQLite local path
- Role-based API protection, cache workers, exports, integrations

Ready to Grow

- Railway/Netlify deployment is documented
- Motive snapshots and route history provide audit base
- Telegram bot extends driver workflow outside browser

NEXT IMPROVEMENTS

- Lock production secrets and CORS for real domains before release.
- Validate real Motive/TomTom credentials and background refresh under load.
- Add smoke tests for auth, loads, route-assistant, motive fleet, fuel-authorizations, safety.
- Keep data dates next to metrics in all statistics and integration blocks.

PDF TAYYOR

To'liq project haqida 10 sahifa